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SUMMARY

ML Engineer and Data Scientist with 3+ years of industry experience building production LLM systems, GenAI data pipelines, and computer vision models. Shipped a GPT-powered news intelligence platform at scale, a full CV pipeline trained on HPC infrastructure, and a LangGraph-orchestrated multi-agent compliance system. Currently building a RAG search engine with BM25, TF-IDF, and semantic retrieval. Comfortable owning the full loop from data ingestion and model training through to serving and evaluation. Actively building toward fine-tuning.

TECHNICAL SKILLS

Languages: Python, R, SQL, Bash
ML & GenAI: PyTorch, scikit-learn, LLM APIs, LangChain, LangGraph, YOLO, OpenCV, DeepFace
RAG & Retrieval: BM25, TF-IDF, semantic search, vector embeddings, FAISS (in progress)
Data Engineering: Apache Airflow, Spark, ETL pipeline design, REST API scraping, OCR, fuzzy matching/entity resolution
Cloud & Infra: AWS S3, Azure, Alibaba Cloud, Docker, Git, SLURM, HPC distributed training, streaming APIs, MLflow
Visualisation: Tableau, Power BI, Quarto, MkDocs

EXPERIENCE

Computer Vision Researcher (Capstone) : UBC Animal Welfare Program — Vancouver, Canada | Apr 2026 to June 2026

- Fine-tuned YOLO on 1,100 annotated video clips across 8 experimental splits to detect cross-sucking behaviour in dairy calves, achieving 0.73 average temporal IoU against a 0.70 baseline in a naturalistic, low-prevalence detection setting.
- Engineered the full pipeline from 300GB of raw barn footage to automated evaluation reports: OpenCV clipping, proximity-based baseline, Seq-NMS temporal linking, and a Makefile workflow integrating SLURM-parallelised HPC training, inference, and Quarto report generation.

Data Scientist II : JLL (Jones Lang LaSalle) — Guangzhou, China | Jul 2023 to Aug 2024

- Architected an end-to-end production ETL pipeline ingesting 9 real estate news outlets via adversarial scraping, orchestrated daily on Alibaba Cloud via Apache Airflow with PolarDB for query serving, replacing a manual weekly research process and delivering city-level news recommendations to external firms in real time
- Built the NLP layer applying GPT-powered summarisation, NER-based location and monetary entity extraction, and cosine similarity deduplication to rank and surface relevant news by city. Added a taxonomy validation layer to catch LLM hallucinations before they reached users, reducing bad outputs in production.

Data Scientist : JLL (Jones Lang LaSalle) — Guangzhou, China | Jan 2023 to Jul 2023

- Led a GPT-3.5 hackathon project: built a Flask platform using Selenium-driven dynamic rendering and runtime LLM-generated code execution to extract structured data from arbitrary websites, enabling zero-code data collection for non-technical analysts
- Designed a multi-signal propensity scoring pipeline fusing corporate signals (job postings, investment flows, policy changes) with scraped POI data to predict client upsizing/downsizing intent, surfacing prioritised leads via a context-aware recommendation system. Contributed to a 30% increase in broker cold-call success rate
- Built a street-view imagery scraper with geo-coordinate extraction to map land plot visual context for an Industry Park white paper, delivering proprietary macro analysis of development pace and land usage ahead of public reporting

Business Data Analyst : JLL (Jones Lang LaSalle) — Guangzhou, China | Sep 2021 to Jan 2023

- Co-authored the Vacancy Anxiety Index: engineered a quarter-aware vacancy tracking algorithm across 4 submarkets, 21 buildings, and 500+ units, combining per-unit destocking duration with an absorption rate model to produce a forward-looking leasing risk indicator used directly in institutional client advisory
- Built IQR-based pricing benchmarks integrating third-party market data to flag statistical outliers in landlord asking rents, translating anomaly detection into premium positioning recommendations for the leasing advisory team
- Automated legacy BI reporting in Python and Tableau and built ad-hoc POI scraping pipelines for new market entry, eliminating manual reporting errors and accelerating regional business development

EDUCATION

Master of Data Science University of British Columbia — Aug 2025 to Jun 2026
MicroMasters: Statistics & Data Science MIT Open Learning — May 2023 to Jan 2025
Bachelor of Commerce: Business Technology Management — Sep 2016 to Jun 2021
Toronto Metropolitan University (formerly Ryerson University)

SELECTED PROJECTS

Reinforcement Learning: Self-Play & Discretized Q-Learning Agents

- Designed and implemented two RL environments and training pipelines from scratch in Python, covering state representation, action space, and reward function design.
- Built a self-play tabular Q-learning agent for tic-tac-toe; diagnosed and fixed a shared-value-function convergence bug, achieving a 0% loss rate against random play across 1,000 evaluation games.
- Implemented discretized Q-learning for CartPole-v1 (Gymnasium) with custom state-space bucketing; iteratively diagnosed training plateaus through reward-curve analysis and hyperparameter experimentation (epsilon decay, discretization granularity).

AI-Driven KYC Onboarding & Compliance Engine

- Architected a three-stage, jurisdiction-aware onboarding pipeline (behavioral fraud gate → OCR/face-match/liveness verification → OSINT compliance agent) as a single generator function reused, unmodified, by both a live streaming API and a CLI batch script, with zero duplicated business logic between the two. Config-driven country routing adds a new jurisdiction with one JSON entry, no new code branch.
- Built a LangGraph-orchestrated compliance agent screening applicants against a real 7,495-entry OFAC sanctions list (fuzzy name/DOB matching), live adverse-media and court-record search, with a local LLM (Ollama) producing a LOW/MEDIUM/HIGH risk summary. Kept the sanctions match deterministic and independent of the LLM output specifically to bound hallucination risk in a compliance-critical decision.

RAG Search Engine

- Building a full-stack retrieval system from scratch implementing BM25 keyword ranking, TF-IDF scoring, inverted index, and semantic search using sentence-transformer embeddings with chunking strategies and hybrid retrieval, served via FastAPI and deployed on Render.